

FINAL PROJECT

Applied Statistics for Engineers and Scientists II

August 5, 2026

Requirements::

- Each group will be responsible for one topic that includes 2 activities. **Groups will work independently on their assigned topics. Copying from other groups is not allowed; the grade will be equally shared among group members.**
- AI tools (e.g., ChatGPT, Gemini, Copilot) may only be used as supportive study aids. All submitted work (including code, data visualizations, result interpretations, and written reports) must reflect the group's own understanding and be articulated in your own words. Directly copying code, tables, interpretations, or report text generated by AI is strictly prohibited and **will result in a zero grade.**
- Each group must submit one report with a maximum length of 60 pages (excluding appendices). The cover page must clearly state the following information: Full names of group members, Student IDs.
- Students are required to use R/RStudio to perform the data analysis in this project.
- The report must include all of the following sections: data collection and cleaning (if applicable), data description, and statistical analysis. Additionally, the report should contain: a table of contents, title, cover image (optional), R code, references, data sources, and a breakdown of individual group member contributions. The report must be presented logically, clearly, and scientifically.
- The report must clearly state the research objectives and significance of the chosen topic, characteristics of the data, statistical methods used, statistical analysis and results, conclusions, recommendations, further discussions, etc.

Submission deadline: August 30, 2026

Each group must upload one compressed file to Moodle, containing:

- 1 PDF file of the report
- 1 folder containing R code files (.R, .html, or .Rmd)
- 1 folder containing the data used for the analysis

Project Topic

Activity 1: (6 points)

Perform Exploratory Data Analysis (EDA), conduct statistical hypothesis testing, and build predictive models to evaluate the factors influencing job salaries in the Data Science field globally. The dataset is retrieved from the "Data Science Job Salaries" dataset on Kaggle, provided in the "ds-salaries.csv" file, consisting of 607 observations and 11 variables:

- "work-year": The year the salary was paid.
- "experience-level": The experience level in the job during the year with the following possible values: EN Entry-level / Junior MI Mid-level / Intermediate SE Senior-level / Expert EX Executive-level / Director
- "Employment-type": The type of employment for the role: PT Part-time FT Full-time CT Contract FL Freelance
- "job-title": The role worked in during the year.
- "salary": The total gross salary amount paid.
- "salary-currency": The currency of the salary paid as an ISO 4217 currency code.
- "salary-in-usd": The salary in USD (FX rate divided by avg. USD rate for the respective year via fxdata.foorilla.com).
- "employee-residence": Employee's primary country of residence in during the work year as an ISO 3166 country code.
- "remote-ratio": The overall amount of work done remotely; possible values are as follows: 0 No remote work (less than 20
- "company-location": The country of the employer's main office or contracting branch as an ISO 3166 country code.
- "company-size": The average number of people that worked for the company during the year: S less than 50 employees (small) M 50 to 250 employees (medium) L more than 250 employees (large)

Using the "ds-salaries.csv" dataset, flexibly apply the statistical and machine learning methods covered in class (e.g., Multiple Linear Regression, Polynomial Regression, Ridge and LASSO Regression, Binary Logistic Regression, Factorial ANOVA) to address the following 03 research questions:

1. Salary Forecasting & Key Drivers Identification: How accurately can we predict the salary (salary-in-usd) of a Data Science professional globally, and which key factors play a crucial role in determining income levels?
2. Impact & Multi-Factor Interaction Analysis: Are there statistically significant differences in salary across experience levels, company sizes, or geographic locations? Do these factors operate independently, or do they exhibit interaction effects with one another in shaping compensation?

3. Top 25% Income Probability Prediction (Top-tier): What specific attributes (e.g., job role, employment type, experience level) increase or decrease the likelihood of a professional entering the top 25% salary bracket, and how effectively can a Logistic Regression model predict this probability?

Task: Students are required to perform the following:

1. Data Preprocessing

- Check for and handle missing values, duplicate records, and potential outliers.
- Group the granular job titles (job-title) into major, meaningful job categories (e.g., Data Analyst, Data Scientist, Data Engineer, Machine Learning/AI Engineer, Management/Lead).
- Encode categorical variables (using Ordinal/One-Hot Encoding) and scale numerical features as necessary.

2. Exploratory Data Analysis (EDA)

- Perform descriptive statistics and visualize the data to understand the distribution of salary-in-usd.
- Visualize the relationships between salary and key factors, including job category, experience level, employment type, remote work ratio, company size, and location.

3. Inferential Statistics & Factorial ANOVA

Select and justify appropriate statistical techniques to:

- Implement a Multi-Factor ANOVA (Factorial ANOVA) model to evaluate mean salary differences across groups while assessing both main effects and interaction effects (e.g., Experience Level $\times$ Company Size, or Experience Level $\times$ Job Category) on salary-in-usd.
- Test the underlying assumptions of ANOVA (normality of residuals, homogeneity of variance) and propose remedy methods if any assumption is violated.
- Analyze trends in salary changes over time (work-year).

4. Continuous Salary Regression Modeling

- Partition the dataset into Training and Testing sets using an appropriate split ratio.
- Construct at least two continuous regression models to predict salary-in-usd (e.g., Multiple Linear Regression, Polynomial Regression, Ridge Regression, LASSO Regression).

5. Binary Logistic Regression Modeling

- Frame a classification problem: Create a binary target variable Y (is-top-tier) based on the 75th percentile (Q_3) of salary-in-usd ($Y = 1$ if the salary falls into the Top 25% highest earning tier, $Y = 0$ otherwise).
- Fit a Binary Logistic Regression model to estimate the probability of a professional reaching the top-tier salary bracket.

- Interpret the estimated coefficients using Odds Ratios ($\exp(\beta)$) to identify key factors that significantly increase or decrease the likelihood of achieving a high salary.

6. Model Evaluation & Comparison

- Evaluate the continuous regression models from step (d) using $RMSE$, MAE , and R^2 metrics.
- Evaluate the Logistic Regression model from step (e) using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Select the best model for each task and justify your choice.

7. Interpretation and Recommendations

- Feature Importance Interpretation: Interpret the significance and impact of key variables in the selected models.
- Propose potential avenues for model improvement or alternative analytical approaches to achieve better predictive performance

Activity 2: (4 points)

Each group must select a dataset on a topic of their choice. Students are encouraged to use real-world data collected from experiments, surveys, or projects within their own major. Alternatively, they may search for datasets from sources such as:

<https://archive.ics.uci.edu/ml/datasets> or <https://www.kaggle.com/datasets>, ...

The dataset must include at least 2 categorical variables and at least 4 numerical variables. Then, complete the following tasks:

1. Describe and clean the dataset (if needed);
2. Perform descriptive statistics on the dataset;
3. Apply appropriate models (studied in this course) to answer the group's proposed research questions:
 - Analyze and interpret the results;
 - Discuss, conclude, and propose recommendations (if any) based on model findings.